Probabilistic Models of the Visual Cortex

Fall 2025 Homework 3

Prof. Alan Yuille

Instructions

1. **Deadline**: Nov 30, 2025 at 23:59:59.
2. **Penalty for late submission**: 20% score deduction per day (or part thereof)
3. **Submission format**: PDF file named <your_jhed>-hw3.pdf.
4. **Warning**: Start the coding part (Question 1.4 and 7) early!
5. **Collaboration**: Feel free to discuss the questions with friends, but be sure to write your answers entirely independently!
6. **LLM Usage**: Do not use LLMs/ChatGPT for looking up answers. You'll receive no credit for this howework!
7. **Questions?** Start a Canvas Discussion, attend office hours, or send an email!

Question 1: Markov Chain (40 points)

A time series x_t is a first order Markov-Chain if

$$p_\theta(x_t \mid x_1, ..., x_{t-1}) = p_\theta(x_t \mid x_{t-1}), \tag{1}$$

i.e. x_t depends on the past values only through x_{t-1}, or, conditional on x_{t-1}, x_t is independent of the past values. An autoregressive ($AR(1)$) model is a Markov model of order one. The simplest but still useful model is a two-state Markov chain, where the data are a dependent series of zeros and ones; for example, $x_t = 1$ if it is raining and zero otherwise. This chain is characterized by a matrix of transition probabilities.

x_{t-1} \ x_t	0	1
0	θ_{00}	θ_{01}
1	θ_{10}	θ_{11}

Table 1: Transition probabilities based on x_t and x_{t-1}.

where $\theta_{ij} = P(X_t = j|X_{t-1} = i)$, for i and j equal to 0 or 1 . The parameters satisfy the constraints $\theta_{00}+\theta_{01} = 1$ and $\theta_{10}+\theta_{11} = 1$, so there are two free parameters. On observing time series data $x_1, ..., x_n$, the likelihood of the parameter $\theta = (\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$ is

$$
\begin{aligned}
L(\theta) &= p_\theta(x_1) \prod_{t=2}^{n} p(x_t \mid x_{t-1}) \\
&= p_\theta(x_1) \prod_{t=2}^{n} \theta_{x_{t-1}0}^{1-x_t} \theta_{x_{t-1}1}^{x_t} \\
&= p_\theta(x_1) \prod_{ij} \theta_{ij}^{n_{ij}} \\
&\equiv L_1(\theta) L_2(\theta) \\
&\equiv L_1(\theta) L_{20}(\theta_{01}) L_{21}(\theta_{11})
\end{aligned}
\tag{2}
$$

where n_{ij} is the number of transitions from state i to state j. For example, if we observe a series

$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$

then $n_{00} = 2, n_{01} = 1, n_{10} = 1$ and $n_{11} = 1$. Again, here it is simpler to consider the conditional likelihood given x_1. The first term $L_1(\theta) = p_\theta(x_1)$ can be derived from the stationary distribution of the Markov chain. The likelihood term $L_2(\theta)$ in effect treats all the pairs of the form (x_{t-1}, x_t) as if they are independent. All pairs with the same x_{t-1} are independent Bernoulli trials with success probability $\theta_{x_{t-1}}$. Conditional on x_1, the free parameters θ_{01} and θ_{11} are orthogonal parameters, allowing separate likelihood analyses via

$$L_{20}(\theta_{01}) = (1 - \theta_{01})^{n_{00}} \theta_{01}^{n_{01}} \tag{3}$$

and

$$L_{21}(\theta_{11}) = (1 - \theta_{11})^{n_{10}} \theta_{11}^{n_{11}} \tag{4}$$

We would use this, for example, in (logistic) regression modelling of a Markov chain. In its simplest structure the conditional MLEs of the parameters are

$$\hat{\theta}_{ij} = \frac{n_{ij}}{n_{i0} + n_{i1}} \tag{5}$$

1. Sadie suffers from allergy attacks during 110 days of autumn. The data (read by row) are the following (0 being no attack):

0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
0	0	0	0	1	1	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0
0	1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 1: Allergy Attack by day

Assuming a first-order Markov model, make the observed transition matrix (like in Table 1). (8 points)

2. What is the estimated probability of an allergy attack tomorrow if Sadie was healthy today? (5 points)

3. What is the probability of an attack tomorrow if Sadie had an allergy attack today? (5 points)

4. Write a python function which can input the above data as a list and outputs the transition matrix as a (2x2) array. Write the function so it can take more than 2 states unlike above data and outputs a variable sized transition matrix. Verify you answer with above data or generate data using *numpy.random.randint* function (whichever you prefer). Your answer should include a screenshot of your data, your code and output in a Jupyter notebook or Google Colab. Also, add a link to your Google Colab notebook (https://research.google.com/colaboratory/faq.html). We will run our data with your code. The input data would be a python list of positive integers. (22

points)

Hint: You could do this part earlier and calculate answers to previous questions in this section.

Question 2: Cue Coupling (5 points)

(Choose any questions which add to 5 or more points)

1. What is the distinction between weak and strong coupling of visual cues? (1 point)

2. Under what conditions can cues be coupled by weighted linear summation? When does this show that cue coupling is optimal in the sense of combining cues taking into account their uncertainty? (1 point)

3. What is double counting, and how can it be avoided? (1 points)

4. Briefly discuss directed graphical models and how they can be applied to model the problem where your friend might have psychic powers. (2 points)

5. Give two examples of strong coupling. (1 points)

6. Briefly describe the multisensory coupling of audio and visual cues, describing how this can be formulated as model selection to determine common cause (or not), and explaining why this formulation can exhibit negative bias. (2 points)

Question 3: Gibbs Sampling (8 points)

(Choose any questions which add to 8 or more points)

The Ising model is specified by a Gibbs Distribution $P(\vec{S}|\vec{I}) = \frac{1}{Z}\exp\{-E(\vec{S};\vec{I})\}$ where the energy $E(\vec{S};\vec{I})$ can be expressed by:

$$E(\vec{S};\vec{I}) = \sum_{x}(S(x) - I(x))^2 + \lambda \sum_{x} \sum_{y \in N(x)} (S(x) - S(y))^2.$$

Here, $N(x)$ denotes the set of pixel indices neighboring x, $S(x) \in \{0, 1\}$ (also called the *state*), and $I(x) \in [0, 1]$ (the *image*).

1. Describe how this model captures spatial context. (1 point)

2. What is the likelihood and prior for this model? (2 points each)

3. Why is it impossible to sample directly from the full joint distribution $P(\vec{S}|\vec{I})$? (2 points).

4. Compared to sampling from the full joint distribution as above, what distribution is sampled from the Gibbs sampling? (2 points)

5. Derive the Gibbs sampling distribution for this model. (4 points)

6. What theoretical results guarantee that Gibbs sampling will converge to samples from the Gibbs distribution? (Note: no need for details or derivations, just broad results). (3 points)

Question 4: Mean Field Theory (15 points)

1. Describe the mean field theory approximation for the Ising model $P(\vec{S}|\vec{I})$. (6 points)

2. What is $Q(\vec{S})$ and how is the Kullback-Leiber divergence used as a measure of similarity between $P(.)$ and $Q(.)$? (6 points)

3. In mean field theory, what is the free energy? (3 points)

Question 5: Boltzmann Machine (7 points)

(Choose any questions which add to 7 or more points)

1. What is the difference between a Boltzmann Machine and a Restricted Boltzmann Machine? (3 points)

2. What is the difference between the hidden and the observed variables $(\vec{S}_o, \vec{S}_h)$. $P(\vec{S}) = \frac{1}{Z}\exp\{-E(\vec{S})\}$, where $E(\vec{S}) = C\sum_{ij}\omega_{ij}S_iS_j$. (4 points)

3. What is the update rule for learning the weights $\{\omega_{ij}\}$ in terms of the expected statistics of S_iS_j with respect to the clamped and unclamped distributions. (4 points)

4. Why is it easier to learn these weights for the Restricted Boltzmann Machine. (3 points)

Question 6: Motion & Kalman Filtering (10 points)

(Choose any questions which add to 10 or more points)

1. List one difference between short-range and long-range motion. (1 point)
2. What is one problem of estimating short-range motion? (1 point)
3. To estimate short-range motion, what are the priors terms and their equations? (4 points)
4. Given the prior, if there is no observation, what are the estimated two dimensional velocities? (2 points)
5. What are the estimated velocities if you have an observation term? (2 points)
6. What are the two stages of Bayes-Kalman filter and their equations? (4 points)
7. What is the purpose of the second step? (1 point)
8. Derive the second step equation by using Bayes' rule. (1 point)

Question 7: Experimental Section: Foreground-background segmentation (16 points)

In this question, you will use Gibbs sampling to apply the Ising model to foreground-background segmentation. Here's a link to the Jupyter notebook with all the code you'll need: `https://github.com/Tiezheng11/ProbabilisticModelsOfVisualCognition2025FA/blob/main/homework_3/gibbs_sampling.ipynb`

For this homework, you do NOT need to submit any code to us. We only expect to see generated plots and answers to the questions in the notebook.

Note: The modeling conventions in the notebook are slightly different from those above. In particular, in the code, S takes values ± 1 and not $\{0, 1\}$.